Private Credit Modeling

Where credit becomes fragile: a listed-BDC market proxy, default modelling on a synthetic loan book, and the arithmetic of redemption gates.

Ajai Upadhyaya

ABSTRACT Private credit is hard to study from the outside: loans are unlisted, marks are quarterly and borrower data is private. This report works with what can be measured honestly. An equal-weight basket of five listed business development companies fell 54.2% from December 2019 to March 2020 and ended March 2026 24.6% below its February 2025 high. On a synthetic 50,000-loan book, two prespecified classifiers rank later-cohort defaults well (holdout AUC 0.877 and 0.882) but overstate their level roughly two- to threefold. A deterministic redemption model shows that a quarterly gate sets how long investors wait, while the realised sale price alone sets how much the remaining investors lose.

01

Summary

The study set out to ask where private credit becomes fragile. It answers three narrower questions, each with a different grade of evidence: how the listed end of the market has traded, how much a default model can learn once its shortcuts are removed, and what redemption gates do to the investors who stay.

−54.2%	**0.882**	**4.02%**	**7.6%**
BDC basket peak-to-trough, 27 Dec 2019 to 23 Mar 2020	Holdout ROC AUC, random forest (95% CI 0.864–0.895)	Expected loss / exposure, severe scenario (base 1.17%)	Loss to remaining investors after a 30% redemption at a 15% haircut

KEY FINDINGS

1. **The listed proxy is volatile and correlated.** The five BDCs (ARCC, FSK, MAIN, OBDC, PSEC) moved together, with an average pairwise daily-return correlation of 0.67, and the equal-weight basket ran an annualised volatility of 25.6%. It compounded at 6.7% a year over the window but closed 24.6% below its high.
2. **Ranking survives, calibration does not.** With outcome-adjacent fields removed and the split made by origination date, both models separate defaulters from non-defaulters (AUC 0.877 and 0.882). Their probabilities are far too high, though: the holdout default rate is 7.7% against 15.5% in training.
3. **Sector carries almost no signal in this book.** Default frequency varies only from 13.2% to 14.8% across ten sectors, even though exposure ranges from 2.4% (Retail) to 18.3% (Financials) of the total.
4. **Gates move time, not loss.** In the redemption model, total haircut loss equals the redeemed amount × (1/realisation − 1), whatever the gate. Redeemers are paid in full, so that loss falls entirely on the investors who stay.

None of the three parts measures the size of the private credit market or its share of corporate financing; Section 02 is explicit about why. The value of the work is methodological. It builds a reproducible pipeline that refuses permissive shortcuts, reports modest results in place of the near-perfect scores an earlier exploratory version produced, and keeps proxy, synthetic and model evidence apart.

02

Scope and grade of evidence

Direct private loans are not traded and their borrowers rarely publish financial statements, so outside researchers usually work through proxies. This report uses three sources, each labelled by what it can support.

Table 1. The three evidence streams and what each can and cannot show.

Stream	Data	Grade	Supports
Market structure	Daily prices for five listed BDCs, 18 Jul 2019 – 24 Mar 2026, 1,680 common trading days	Real, unverified snapshot	How listed private-credit vehicles have been priced. Not loan valuations, and not a verified total-return series.
Default modelling	50,000 loans, originations Jan 2015 – Dec 2023, 10 sectors, $164.9bn exposure	Synthetic	Whether a disciplined pipeline recovers signal without leakage. Not real-world predictive power.
Liquidity	Deterministic redemption model; no data	Model	The accounting of gates and fire-sale haircuts under stated assumptions. Not a forecast of any fund.

An earlier version of the project also pulled a macroeconomic table (high-yield spreads, base rates, delinquencies). An audit found that its loader silently substituted synthetic values whenever the public data request failed, so that table is excluded from every result here.

03

The listed end of the market

Business development companies are closed-end vehicles that lend to middle-market firms and trade on public exchanges. Their share prices are a daily, if noisy, reading of how investors price private-credit risk, while the loans themselves are marked quarterly. The basket below is rebalanced daily to equal weights across ARCC, FSK, MAIN, OBDC and PSEC, uses only dates on which all five traded, and is rebased to 100. Recomputing it from the bundled price snapshot reproduces the stored results exactly: 1,680 observations, a maximum drawdown of −0.5415 and a closing level of 153.89.



Figure 1. Equal-weight BDC basket (top) and drawdown from its running peak (bottom), daily. The deepest fall, −54.2%, ran from 27 Dec 2019 to 23 Mar 2020; the basket regained its prior peak on 10 Feb 2021.

Source: privatecredit/data/db/privatecredit.db, table bdc_prices; method of scripts/build_portfolio.py, recomputed.

Three episodes stand out. The March 2020 sell-off took roughly 11 months to recover. A long rise followed, with a gain of 43.8% in 2021 alone, to an all-time high of 204.0 on 19 February 2025. From there the basket slid to 153.9 by 24 March 2026, a drawdown of 24.6% that was still open at the end of the sample. Over the whole window the basket compounded at 6.7% a year.

Table 2. Constituents over the common window. Changes are in the snapshot's price series.

Ticker	Change, Jul 2019 – Mar 2026	Max drawdown	Date of trough	Annualised volatility
ARCC	+87.7%	−56.8%	23 Mar 2020	29.9%
FSK	+11.2%	−60.3%	23 Mar 2020	30.7%
MAIN	+116.6%	−64.5%	23 Mar 2020	31.5%
OBDC	+45.5%	−56.1%	23 Mar 2020	27.1%
PSEC	−12.7%	−55.2%	23 Dec 2025	30.0%
Basket	+53.9%	−54.2%	23 Mar 2020	25.6%

Volatility: standard deviation of daily returns × √252. Average pairwise correlation of daily returns 0.67 (minimum 0.53).

Two details matter for interpretation. First, diversification inside the sector is limited: the basket's volatility (25.6%) is only modestly below that of its members (27–32%), because the names fall together when credit sentiment turns. Second, the members diverge sharply over longer horizons. MAIN more than doubled while PSEC lost ground and recorded its deepest drawdown not in 2020 but in December 2025. Over the long run, the choice of manager mattered as much as the sector, and a basket averages that away.

Table 3. Basket return by calendar year.

2019*	2020	2021	2022	2023	2024	2025	2026*
+9.2%	−9.5%	+43.8%	−6.9%	+23.7%	+16.7%	−9.5%	−10.9%

* Partial years: from 18 Jul 2019; to 24 Mar 2026.

> PROVENANCE
>
> The prices come from a bundled database snapshot whose fetch dates and adjustment settings were not recorded. The series may or may not include reinvested distributions, which are large for BDCs, so these figures are not a verified total-return backtest. A fixed five-name basket chosen today also carries survivorship bias.

04

Default modelling without the shortcuts

The loan book has 50,000 synthetic corporate and commercial loans originated between January 2015 and December 2023, with $164.9bn of exposure at default and a lifetime default flag set on 13.9% of loans. The data are described as an educational dataset of Kaggle origin; the original listing and its licence were not recorded, so raw rows are not redistributed.

Design

An earlier exploratory pipeline reported test AUCs between 0.988 and 1.000. An audit traced those scores to leakage: the models could see fields computed from the outcome, such as the dataset's own expected loss, risk-weighted assets, survival months and recovery rate. The rebuilt study follows four rules.

- **Chronological split.** Loans originated before 1 March 2022 train (39,897 loans); later cohorts form the holdout (10,103 loans). The two sets share no loans.
- **Pre-outcome allowlist.** The 11 features are credit score, exposure, coupon, leverage, interest coverage, debt-to-equity, maturity, sector, loan type, collateral and initial rating. Nine outcome-adjacent fields are excluded, including the dataset-supplied PD and LGD.
- **Training-only preprocessing.** Median imputation and scaling for numeric fields, and most-frequent imputation and one-hot encoding for categorical fields, are fitted on the training cohort alone.
- **Prespecified models.** The two models, a logistic regression ($C = 1$) and a random forest (200 trees, depth 7, minimum leaf 30), were fixed in advance and not tuned on the holdout. Both are reported.

Table 4. Holdout performance, 10,103 loans originated from 1 March 2022 (default rate 7.73%).

Model	ROC AUC	95% CI	Average precision	Brier	Log loss
Logistic regression	0.8768	0.859–0.891	0.425	0.0754	0.280
Random forest	0.8819	0.864–0.895	0.439	0.0649	0.257
Constant baseline	—	—	—	0.0773	—

CI: 200 bootstrap resamples of the holdout. Baseline predicts the training default rate for every loan.

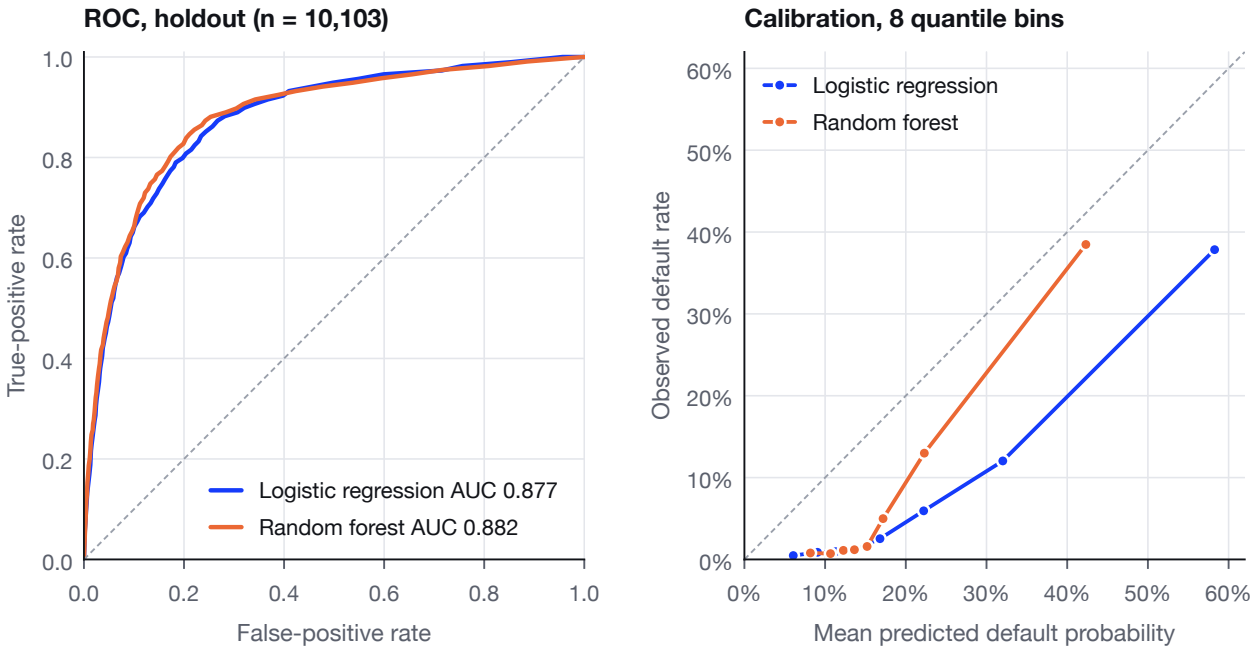


Figure 2. Left: ROC curves on the holdout. Right: mean predicted probability against observed default rate in eight equal-count bins; points below the dashed line are over-predictions.

Source: privatecredit/dist/research.json (credit.models[].roc, .calibration), produced by the companion creditrisk pipeline.

Reading the results

Both models rank risk well. Average precision of 0.43–0.44 is 5.5–5.7 times the 7.7% prevalence, and the two AUC intervals overlap. The probabilities are another matter. Averaged across equal-count bins, the logistic model predicts about 21% default and the random forest about 18%, against 7.7% observed. In the safest bin the logistic model predicts 6.1% and sees 0.5%. In the riskiest bin it predicts 58.3% and sees 37.9%. The random forest's top bin is close (42.3% predicted, 38.5% observed), but its lower bins overpredict tenfold.

The Brier scores tell the same story. The random forest improves on the constant baseline by 16%, while the logistic model improves by only 2.5%: its ranking skill is almost cancelled by miscalibration. The cause is a shift in the base rate, from 15.5% in the training cohorts to 7.7% in the holdout. Loans originated after March 2022 have had less time to default. Because outcome timestamps and observation horizons are not validated in the dataset, that reading is consistent with the data but not proven.

> IMPLICATION
>
> A model that ranks well can still misprice. Any use of these scores for loss budgeting would need recalibration on a horizon-matched target, such as a twelve-month default flag, not the lifetime flag supplied here.

05

Concentration and stress

Exposure is unevenly spread across the ten sectors. Financials, Real Estate and Utilities hold 47.0% of it, and the Herfindahl index of sector shares is 0.124, against 0.100 for an even split. Default frequency is almost flat across sectors. In this book, sector is a concentration fact, not a risk signal.

Table 5. Sector concentration, full book (50,000 loans).

Sector	Loans	Exposure ($bn)	Share of exposure	Default frequency
Financials	5,023	30.17	18.29%	14.75%
Real Estate	4,862	25.31	15.34%	13.94%
Utilities	4,959	22.07	13.38%	13.35%
Energy	5,132	20.94	12.69%	14.46%
Telecom	5,077	18.91	11.47%	13.49%
Industrials	5,012	15.71	9.53%	14.41%
Technology	4,959	12.77	7.74%	13.63%
Healthcare	4,982	10.05	6.09%	13.19%
Consumer	5,043	5.05	3.06%	13.70%
Retail	4,951	3.96	2.40%	14.06%

Expected-loss stress

The stress test runs separately from the classifiers. It applies multiplicative shocks to the dataset's annual PD and additive shocks to its LGD, caps both at 100%, and sums exposure × PD × LGD across the book.

$$EL = \Sigma_i \, EAD_i \cdot min(1, \, m \cdot PD_i) \cdot min(1, \, LGD_i + a) \tag{1}$$

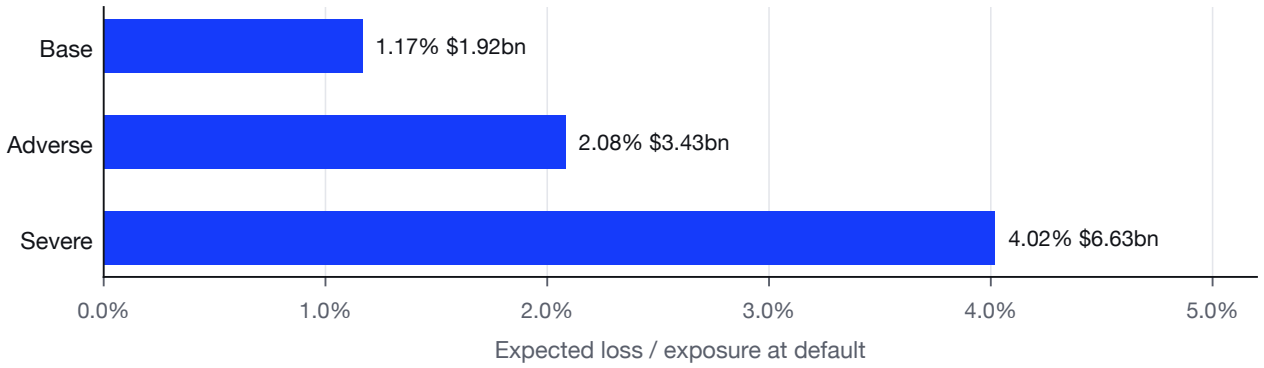


Figure 3. Expected loss as a share of $164.9bn exposure, by scenario: Base (m = 1.0, a = 0), Adverse (1.5, +0.10), Severe (2.5, +0.20).

Source: privatecredit/dist/research.json (credit.stress).

Expected loss rises from $1.92bn (1.17% of exposure) in the base case to $3.43bn (2.08%) under the adverse scenario and $6.63bn (4.02%) under the severe one. That is 3.4 times the base case for a 2.5× PD shock: the LGD add-on compounds the PD multiplier. These shocks are illustrative assumptions, not calibrated macro scenarios, and the PD and LGD inputs are themselves synthetic. The 1.17% base figure is an annual expected loss. It is not comparable with the 13.9% lifetime default flag used for classification.

06

The arithmetic of redemption gates

Semi-liquid private credit funds promise periodic liquidity on assets that cannot be sold quickly at book value. The model here isolates that mismatch. A fund with initial NAV A_0 faces requests for a fraction R of NAV. Each quarter it pays out at most the gate, a fraction g of initial NAV, and funds the payment by selling assets that realise a fraction ρ of book value.

$$paid_t = min(pending_t,\ g{\cdot}A_0,\ \rho{\cdot}A_t) \qquad sold_t = paid_t / \rho \qquad loss_t = sold_t - paid_t \tag{2}$$

The model assumes no income, borrowing, new requests or price recovery. When assets run out, unfunded requests stay outstanding instead of disappearing. All amounts are in units of initial NAV = 100.

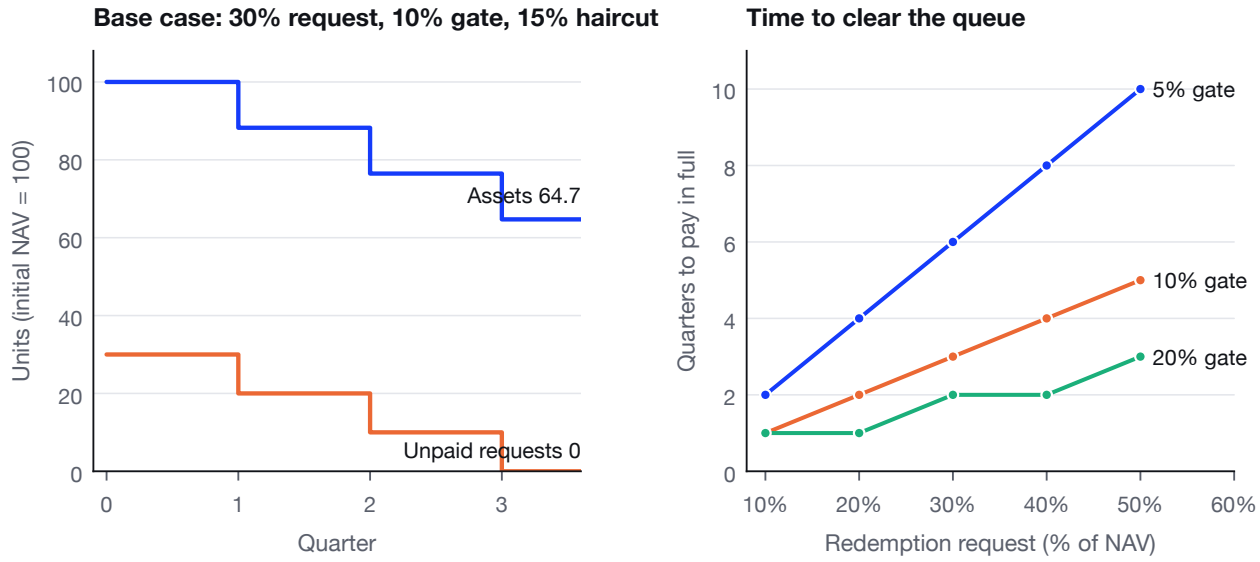


Figure 4. Left: the base case, with a 30% request, a 10% gate and 85% realisation. Right: quarters needed to clear requests of 10–50% of NAV under 5%, 10% and 20% gates, at the same realisation.

Source: privatecredit/src/models/liquidity_stress.py, run for this report; the base case matches research.json (liquidity).

Table 6. Base case by quarter (R = 30%, g = 10%, ρ = 0.85).

Quarter	Paid	Assets sold	Haircut loss	Requests outstanding	Assets remaining
1	10.00	11.76	1.76	20.00	88.24
2	10.00	11.76	1.76	10.00	76.47
3	10.00	11.76	1.76	0.00	64.71
Total	30.00	35.29	5.29		

Two results that follow from the accounting

The gate sets the pace; the haircut sets the cost. While assets suffice, the redeemed amount $R{\cdot}A_0$ is paid in full, and every unit paid costs $1/\rho$ units of assets. Total haircut loss is therefore $R{\cdot}A_0{\cdot}(1/\rho - 1)$ whatever the gate. Across every gate and request size in Figure 4 the simulated loss matches this identity to machine precision. The gate changes only the wait, which is about R/g quarters. A 50% request takes ten quarters under a 5% gate and three under a 20% gate.

The cost lands on those who stay. Redeeming investors receive 30.00 for claims of 30.00. The 70 units of NAV that remain are now backed by 64.71 of assets, so investors who stayed lose 7.6% while those who left lose nothing. That is a first-mover advantage built into the accounting, and it does not depend on the gate. For a 30% request, the stayers' loss is 1.6 units at 95% realisation, 3.3 at 90%, 7.5 at 80% and 12.9 at 70%.

> SCOPE
>
> This is mechanics, not a forecast. Real funds earn income, can borrow, sell their most liquid assets first, and face requests that respond to each other. Each of those would change the numbers but not the direction of the two results above.

07

Limitations

- **No direct private-credit data.** The study has no loan-level direct-lending data, no PIK or non-accrual records, no fund NAV marks or spreads, and no borrower financials. It cannot speak to the size of the market or its role in financing firms.
- **Proxy prices are unverified.** Fetch dates and dividend adjustment for the BDC snapshot are unknown. The five names are a fixed, present-day selection.
- **Loan outcomes are synthetic.** Good discrimination on generated data does not establish real-world predictive ability, and the dataset's licence is unrecorded.
- **The target is a lifetime flag.** Loans have different lifetimes and unvalidated observation windows, which likely drives the training-to-holdout base-rate shift and the miscalibration in Section 04.
- **One split, one seed.** The bootstrap intervals capture holdout sampling only. They do not cover variation across splits or model specifications.
- **Stress inputs are circular.** Expected loss is computed from the dataset's own synthetic PD and LGD, under shocks chosen for illustration.

08

What a firm-financing study would need

The question the project began with is how much firms now rely on private lenders, and how fragile that channel is. Answering it would take data this repository does not hold: loan-level schedules of investments as BDCs disclose them in periodic filings, with spreads, PIK share and non-accruals over time; NAV and redemption histories for semi-liquid funds; and a verified total-return price source. The pipeline here is ready for that data. It validates inputs, splits by time, keeps features to an allowlist, and fixes its models before seeing the holdout.

A

Data and reproducibility

REPOSITORY	ajaiupadhyaya/privatecredit (“Credit Observatory”), commit c4a0b85, 29 Oct 2026. Loan-level analysis from the companion repository ajaiupadhyaya/creditrisk, commit 214f597.
MARKET DATA	data/db/privatecredit.db, table bdc_prices (SHA-256 ad306ff5…fe7309). Common window 18 Jul 2019 – 24 Mar 2026, 1,680 observations.
LOAN DATA	loan_portfolio.csv, 50,000 rows (SHA-256 129dbdd3…f806e). Synthetic; raw rows not redistributed.
RESULTS FILE	dist/research.json, which holds the market series, model metrics, ROC and calibration points, sectors, stress and the liquidity base case.
ENVIRONMENT	Python 3.11.15, pandas 2.3.2, numpy 1.26.4, scikit-learn 1.3.0.
VERIFICATION	The repositories report 10 Python and 7 Node tests for the site and model, and 4 tests for the credit pipeline. The full credit run was repeated twice with matching hashes on six outputs.
THIS REPORT	All four figures were redrawn from the files above. The basket was recomputed from the price snapshot and asserted equal to the stored drawdown and closing level. The liquidity grid was produced by running the repository's own model. Per-ticker statistics, calendar returns, correlations, the Herfindahl index, Brier improvements and stayer losses are derived here from the same files.

Ajai Upadhyaya · esnupi Publications No. 01 · October 2026 · Set in Charter, Archivo Black and Courier New.